

A Dual-Track Cloud–Edge Digital Twin

for Gas Turbine Predictive Maintenance

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PROJECT CONTEXT & MOTIVATION

In the oil and gas industry, aeroderivative gas turbines operate under extreme thermodynamic stress, where undetected degradation in components like the High-Pressure Turbine (HPT) can lead to catastrophic failure. Traditional maintenance is strictly schedule-based, leading to unnecessary downtime. Furthermore, modeling high-frequency thermodynamic deterioration (e.g., NASA N-CMAPSS data) using standard deep learning (like LSTMs) results in severe computational bottlenecks and a failure to capture complex, non-linear degradation physics across different turbine fleets.

CORE OBJECTIVE

This thesis proposes the construction of a proactive **Gas Turbine Digital Twin** to predict the Remaining Useful Life (RUL) and detect early anomalies. By hybridizing lightning-fast state-space sequence models with heavy physics-aware cloud architectures, the system achieves real-time, zero-latency inference on industrial edge devices while maintaining deep generalization capabilities across heterogeneous fleets.

ARCHITECTURAL CONTRIBUTIONS

The project successfully engineers three distinct, novel algorithmic architectures to overcome both computational and physical limitations:

- **Mamba-SSM (Edge Tier):** A hardware-aware state space model that achieves linear-time $\mathcal{O}(L)$ scaling. Engineered specifically for low-compute Programmable Logic Controllers (PLCs), it utilizes a SiLU Gating Branch to filter sensor noise, acting as a “smart valve” to provide real-time degradation indices and RUL tracking directly on the asset.
- **ST-KDFormer (Cloud Tier):** A Spatio-Temporal Koopman-Distributed Transformer. It utilizes Koopman Operator theory to lift the non-stationary, non-linear thermodynamic deterioration into an infinite-dimensional linear space. Crucially, it monitors the Koopman Spectral Radius (ρ), serving as a mathematical certificate of stability to guarantee safe turbine operation.
- **MAML Fast Adaptation (Fleet Generalization):** Standard retraining to a new fleet takes hours of GPU compute. Model-Agnostic Meta-Learning (MAML) solves this by optimizing the initialization weights. Using just a small support set of sensor readings, it adapts the cloud model to an entirely new failure mode (e.g., from Fan degradation to HPT erosion) in **under 1.2 seconds** via a few inner gradient steps.

DATA INTEGRITY & BENCHMARKING

The models were trained on the massive, high-fidelity **NASA N-CMAPSS** dataset. To prevent the “Initial Degradation Paradox” (where short-lived engines start at a low health percentage), the model was trained to predict a normalized Health Degradation Index. In production, this index is mathematically scaled by the manufacturer’s nominal lifespan. Against robust baselines, the hybrid approach drastically reduced Root Mean Square Error (RMSE), achieving state-of-the-art results for RUL prediction.

EXPLAINABLE AI (XAI) & DEPLOYMENT

Because plant operators cannot shut down a multi-million dollar turbine based on a black-box alert, the system utilizes a multi-modal XAI strategy. We employed **SHAP values** and **Integrated Gradients** to isolate causal physical parameters, distinguishing between compressor fouling and turbine erosion.

The entire framework is deployed as a real-time, interactive **Next.js / React Three Fiber 3D Dashboard**, backed by a **FastAPI** engine featuring a specialized Retrieval-Augmented Generation (RAG) AI Copilot to interact directly with engineering manuals.

CONCLUSION

This research proves that a Dual-Track Edge-Cloud architecture embedded with Koopman dynamics and Meta-Learning is the definitive solution for predictive maintenance, achieving both industrial-scale computational latency and indispensable causal transparency for field engineers.